

Docker Desktop Setup & WSL 2 Kernel Issue Resolution

End-to-end diagnostic workflow, WSL 2 kernel remediation, and container runtime validation on Windows 10.

OPERATING SYSTEM	DOCKER DESKTOP VERSION	BACKEND SUBSYSTEM	RESOLUTION STATUS
Windows 10 x64 (AMD64)	v29.7.2 (API v1.55)	WSL 2 Engine	Operational / Verified

1 Initial Symptom & Diagnostic (CLI Verification)

DAEMON OFFLINE

Following initial Docker Desktop installation, running `docker version` in PowerShell returned only the **Client** metadata. The **Server (Docker Engine)** section was absent, confirming that the daemon process was not running.

```
PS C:\Users\Siva Ponakala> docker version
Client:
Version:      29.7.2
API version:  1.55
Go version:   go1.26.5
OS/Arch:      windows/amd64
Context:      default
# Note: Server section was not returned (Docker daemon inactive)
```

2 Docker Desktop Initialization & Terms Acceptance

GUI SETUP

Docker Desktop was opened to initialize the GUI client and start background service daemons.

- **Action:** Clicked **Accept** on the *Docker Subscription Service Agreement* dialog to complete the initial setup sequence.
- **Behavior:** The application initiated subsystem health checks, probing the underlying virtualization engine.

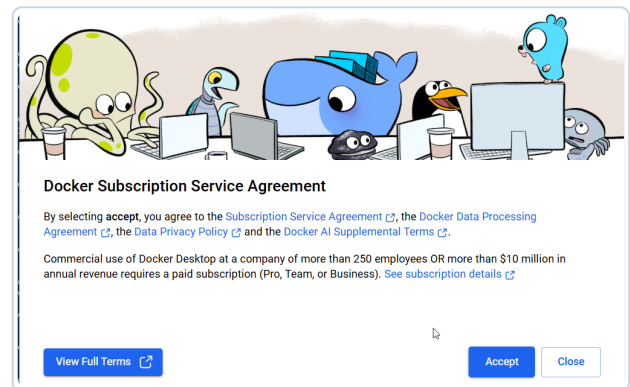


Figure 1: Initial Docker Service Agreement prompt

3 Root Cause Identification & WSL 2 Kernel Remediation

WSL OUTDATED

Root Cause: Docker Desktop requires an updated Windows Subsystem for Linux (WSL 2) kernel package to run Linux container virtualization. An outdated WSL kernel prevented engine initialization.

Resolution Steps (PowerShell Administrator):

```
# 1. Update WSL Linux kernel to latest release
PS C:\WINDOWS\system32> wsl --update

# 2. Restart WSL Subsystem
PS C:\WINDOWS\system32> wsl --shutdown
```

After completing the update, clicking **Try Again** in the modal re-triggered the Docker Engine startup cleanly.

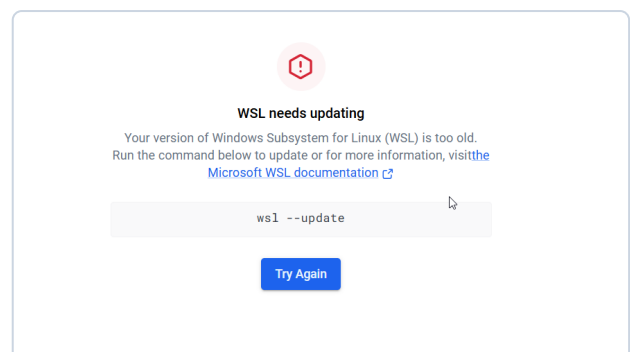


Figure 2: WSL kernel update requirement dialog

4 Docker Desktop Active Dashboard

With the WSL kernel updated, Docker Desktop launched into an active, healthy state.

- **Containers View:** The main dashboard displayed *"Your running containers show up here"*.
- **Engine Status:** Background Docker daemon connected to WSL 2 and ready to accept container operations.

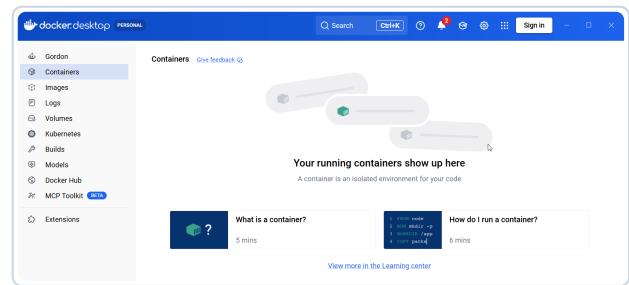


Figure 3: Docker Desktop successfully running

5 End-to-End Runtime Validation (Hello-World)

A live container execution test was conducted via PowerShell to validate complete client-to-engine communication:

```
PS C:\WINDOWS\system32> docker run hello-world
Unable to find image 'hello-world:latest' locally
latest: Pulling from library/hello-world
4f55086f7dd0: Pull complete
Digest: sha256:5dd0d3e6e255913fc30f90b9f2b1d359cc2cbdb48090cc4b65f1676e203243cc
Status: Downloaded newer image for hello-world:latest
```

```
Hello from Docker!
This message shows that your installation appears to be working correctly.
```

Key Troubleshooting Takeaways

1. Missing Server Metadata: If docker version omits the Server section, the Docker daemon is offline or blocked by a virtualization dependency.

2. WSL 2 Kernel Lifecycle: Running `wsl --update` and `wsl --shutdown` fixes common kernel desynchronization issues during Docker Desktop startup.

3. Full-Stack Confirmation: Successfully executing `docker run hello-world` proves network pull, container isolation, execution, and stdout streaming.